Last time:

$$2x_1 + 3x_2 = 10$$
$$4x_1 + 11x_2 = 16$$

We discussed how to solve this, but let us now look at it from another perspective. Same fundamental ideas but quite different notationally.

Write this as a matrix equation:

$$\begin{bmatrix} 2 & 3 \\ 4 & 11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 16 \end{bmatrix}$$

coefficient matrix

Maybe "work on coefficient matrix to solve system? Problem is that doesn't take into account right side of the equation so doesn't fully represent the system.

So more useful maybe to look at the <u>augmented matrix:</u>

$$\begin{bmatrix} 2 & 3 & 10 \\ 4 & 11 & 16 \end{bmatrix}$$

which has the values on right side which you wish to equal as an "appended" column. So maybe can work on this instead since this now has the full information of the system? But how?

We use a very important idea called <u>elementary row operations</u> (row operations in short).

These are used in many contexts to "simplify" a matrix and thereby hopefully make it easier to work with.

Solving linear systems is only one such context (though certainly probably the most common context).

Elementary Row Operations:

(1) Replace a row by itself plus multiple of another row
(2) Interchange two rows
(3) Multiply a row by a nonzero constant

Two systems whose augmented matrices are <u>row equivalent</u> (equivalent in short) are the same system and have the same solutions.

Why do these make sense (at least in context of solving a system)? Well the first one, adding a constant multiple of one equation to another is a very common trick to eliminate a variable - that's same thing as that row operation is the key. Other two similar.

When system has same number of equations as variables, like the 2 x 2 system earlier, if we don't check explicitly, our suspicion would be there will be a

unique solution, but ok not guaranteed.

If number of equations and variables are not same, we have more reason to suspect no solutions or many solutions. To that end, two big questions are:

(1) Is the system <u>consistent</u>, i.e. is there at least one solution?
(2) If system is consistent, is there a unique solution or many solutions?

Say we are solving a system which is 10 x 10, 10 x 9, or 9 x 10. While it might not be practical to solve it, might we be able to answer these questions without explicitly solving it?

And if that too is not practical how would we try at least?

Let us first try to assess that in the situation where the number of variables is the same as number of equations, i.e. coefficient matrix is a square matrix.

Here it is often useful to put matrix in <u>triangular form.</u>

For a square matrix (well any matrix, but mostly used in square matrices), the entries which are in same row as column, i.e. first row + first column, second row + second column are the <u>main diagonal</u> or diagonal for short. E.g. for a 4 x 4 matrix:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

Blue is diagonal. A square matrix is <u>upper triangular</u> if every entry below diagonal is 0.

<u>Lower triangular</u> all above diagonal are 0. They are similar in usefulness so for consistency we mostly use upper.

Why in a linear system is it useful for coefficient matrix to be triangular? E.g.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 7 & 8 \\ 0 & 0 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -5 \\ 7 \\ 100 \end{bmatrix}$$

Equations from bottom to top are:

$$9x_3 = 100 \quad \text{---->} \quad \text{get } x_3$$
$$7x_2 + 8x_3 = 7 \quad \text{get } x_2$$
$$x_1 + 2x_2 + 3x_3 = -5 \quad \text{get } x_1$$

So this isn't much work to solve really, no elimination of variables or putting together equations needed.

Even if system was 10 x 10, if coefficient matrix is triangular, not so bad to solve using same idea, far easier than if matrix was "generic", i.e. not triangular.

So motivation to have square matrix be triangular is clear, but how do we go about changing it into triangular if it is not already so?

<u>Ex:</u>

$$\begin{bmatrix} 1 & 2 & 3 \\ 6 & 5 & 4 \\ 7 & 8 & 9 \end{bmatrix}$$

Colored entries must become 0. First turn blue into 0, that is entire first column other than top entry, via elementary row operations.

Replace R2 with R2 - 6R1, also R3 with by R3 - 7R1.

Be careful when doing multiple operations at once.

If we instead replaced R3 by something involving R2, would that mean starting R2 or the updated one which comes from the first alteration? That's the problem, it'd be very easy to make a mistake there if not careful.

This gives the row equivalent:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -7 & -14 \\ 0 & -6 & -12 \end{bmatrix}$$

Replace R2 with $\frac{-1}{7}$ R2 and R3 with $\frac{-1}{6}$ R3 to get:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -1 & 2 \end{bmatrix}$$

Finally replace R3 with R3 - R2 to yield:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Now is upper triangular.

Important: do not try after making first column 0 (aside from top entry) to replace third row by itself plus three times first! That makes green circled entry 0, but the entry to the left of that won't be 0 anymore! Must not alter what you previously accomplished, that's a very common way that people slip up.

Only work with rows beyond the first row after fixing the first column, that will ensure the 0s in the first column do not get disturbed. And similarly after fixing second column only work with rows beyond the first two.

1.2 Echelon Forms

Say we have a 2 x 100 matrix, e.g.

$$\begin{bmatrix} 1 & 3 & 5 & 7 & \dots \\ 2 & 4 & 6 & 8 & \dots \end{bmatrix}$$

Diagonal blue again. To make this matrix "upper triangular" would only have to make the red entry 0, no problem.

But lower triangular would need to make 197 entries equal 0! Whether possible or not, that's very different! Hence we don't really want to use the word triangular with nonsquare matrices as the lack of similarity between upper and lower makes the term less useful.

The counterpart term we use for nonsquare matrices is <u>row echelon form</u> (abbreviated REF). Requirements are:

(1) All zero rows (zero row means every entry 0) must be at the bottom of the matrix.

(2) The leading entry in each row (the first nonzero entry if there is one) must be to the right, i.e. in a later column, of the row above it (must be later, tied is not allowed).

Really the second condition is the one needed, the first one can be viewed to follow from that, perhaps only in a vacuous sense (hence why we list it separately just to be extra careful).

We put a matrix in REF same way we make square matrix upper triangular, going left to right, column by column.

The earlier square matrix which we made upper triangular is in REF, so those ideas are quite similar but not equivalent.

We also have the <u>reduced row echelon form</u> (RREF) which adds two additional conditions to REF and very importantly is unique for each matrix unlike REF.

(3) All leading entries are 1.
(4) All leading entries have only 0s in the rest of their column.

Leading entries being 1 is simple as we can, at an appropriate moment, multiply each row by a needed constant to make the leading entry such. The other condition is the harder one to bring about.

How to do that? Related to that are the <u>pivot positions</u> of a matrix. A spot in a matrix is a pivot if it is the location of a leading 1 in the RREF of a matrix (one immediate reason the uniqueness of the RREF helps).

A column which contains a pivot position is a <u>pivot column</u>.

Could also talk about pivot rows, but in RREF the zero rows are always at bottom so top rows are always pivot rows while bottom ones (should there be zero rows) are not pivot rows. Pivot columns can vary in location far more, making them more interesting.

Very important to only try to assess a matrix's pivot positions if you have it in REF, leads to trouble if not.

For example:

$$\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$$

Is the bottom right entry a pivot position? Well you might think so since it's first nonzero entry in that row.

But putting this matrix in REF, replacing R2 by R2 - 2R1,

$$\sim \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

From this (actually the RREF), the bottom right is NOT a pivot position. That's why it's dangerous to try to draw conclusions about pivots without REF.

Another example:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Is top left entry not a pivot position since it's 0? Again that might be initial instinct, but putting matrix in REF via switching rows we get

$$\sim \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

It is now clear top left is a pivot position.

So it's not about the value of the entry. Only reliable if matrix is in REF and even then, if it is nonzero, we can multiply the row by a nonzero constant to make that entry anything nonzero so again not about the value of the entry, it's about locations which don't change under elementary row operations due to the uniqueness of the RREF. The actual values of the entries can change though.

How to change from REF to RREF? E.g.

$$\begin{bmatrix} 2 & 3 & 4 & 6 & 7 & 2 \\ 0 & 5 & 5 & 3 & 8 & 5 \\ 0 & 0 & 1 & 5 & 6 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Blue are pivots and red must become 0 as part of putting matrix in RREF. How to bring this about?

Take lowest pivot and make its column all 0s (other than the pivot entry itself). Here replace R3 by R3 - 6R4, R2 by R2 - 8R4, and R1 by R1 - 7R4 to give:

$$\begin{bmatrix} 2 & 3 & 4 & 6 & 0 & -5 \\ 0 & 5 & 5 & 3 & 0 & -3 \\ 0 & 0 & 1 & 5 & 0 & -4 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Now turn the next lowest pivot's column to 0s by replacing R2 with R2 - 5R3 and R1 with R1 - 4R3, get:

$$\begin{bmatrix} 2 & 3 & 0 & -14 & 0 & 11 \\ 0 & 5 & 0 & -22 & 0 & 17 \\ 0 & 0 & 1 & 5 & 0 & -4 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Then make final red entry 0, after that leading entries turn them all into 1 via scaling rows, then done.

This is the process to follow to create RREF from REF.

Key is: entries made 0 in pivot columns prior will stay 0 with later operations for same reason that 0s made 0 when we make square matrix upper triangular stay as such. And since leading entries have 0s below them, they don't get disturbed so matrix stays in REF.

One big reason pivots help us relates to the question we asked about consistency/number of solutions.

<u>Existence and Uniqueness Theorem:</u>

If a linear system is represented by an augmented matrix, that system is consistent if and only if that matrix has no pivot in the last column.

And when consistent, if every prior column has a pivot, then solution is unique, otherwise infinite solutions.

Why? Well if last column has a pivot, in RREF the row with that column's pivot has equation which reads as:

$$0x_1 + 0x_2 + 0x_3 + \ldots + 0x_n = 1$$

last being 1 since pivot position in RREF. And since this obviously has no solution, system is inconsistent.

Once consistent if every prior column has pivot, each variable is forced one by one, similar to upper triangular matrix we did earlier where every diagonal is nonzero, which makes the solution unique.

So since pivots are important do we really need RREF because that form is in theory how we find them?

In general when solving things where REF/RREF is needed people often naturally ask if going to REF is enough or whether they need RREF.

Depends on question, but in this class most of the time REF is enough. Why?

(1) RREF makes systems significantly easier to solve in large systems, in small systems it doesn't make much difference. And since practically you won't be given large systems, not much help there.

(2) The uniqueness of the RREF is very helpful when trying to prove things. But since you won't be doing many proofs in this class, that uniqueness isn't hugely useful to you.

(3) The things which we use RREF for, usually REF does as well. For instance with pivots, the leading entries in a REF form are the pivots still, just as in RREF. So we can find pivots (determine consistency) and number of solutions via REF, we don't need RREF.

So while RREF makes things easier to work with/solve than a generic REF, the extra work required to change from REF to RREF (which can be quite a bit!) is usually not worth the tradeoff, in part for the reasons above.

In a consistent system, looking at the augmented matrix, in the columns other than last one, a nonpivot column corresponds to a <u>free variable</u> while a pivot column corresponds to a <u>basic variable</u>.

For instance using earlier matrix (assuming that is representing an augmented matrix):

$$\begin{bmatrix} 2 & 3 & 4 & 6 & 7 & 2 \\ 0 & 5 & 5 & 3 & 8 & 5 \\ 0 & 0 & 1 & 5 & 6 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

In this, the two lower equations are:

$$x_5 = 1$$
$$x_3 + 5x_4 + 6x_5 = 2$$

x_5 is forced, natural since pivot column, basic variable. x_4 is not forced, also natural since nonpivot column, free variable. x_3 is then forced once x_4 is picked.

So even though x_3 can take different values unlike x_5, it is forced by the prior variables so it is a basic variable, as it should be since pivot column.

Likewise x_1, x_2 are forced by x_4 so overall solution has one free variable.

Hence since that free variable can take any value, there will be infinitely many solutions.

So therefore linear systems always have 0, 1, or infinite solutions, because once consistent, solution is either unique or has free variable(s) so infinite, i.e. linear system can't have say exactly ten solutions.

In next sections we will learn better how to express the solutions to a linear system which has infinite solutions in a convenient way.